

# Cluster-Based Student Profiling and Explainable Early Engagement Prediction Using OULAD

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## 1. SCIENTIFIC BACKGROUND

The Open University Learning Analytics Dataset (OULAD) records 32,593 student module registrations across seven STEM courses delivered in 2013-2014, which provides VLE clickstream logs, demographic attributes, and assessment records. As a widely adopted benchmark in learning analytics research, OULAD offers a rich basis for studying student disengagement in distance education (Kuzilek et al., 2017). However, approximately 32% of those students received a final outcome of Withdrawn, indicating a failure to complete the module. Furthermore, a large proportion of these withdrawals occur during the early stages of a student's first module, before any formal assessment signal can alert institutional staff, leaving a critically narrow window for timely intervention. To address this early disengagement problem, this project aims to construct interpretable student behavioural profiles via k-Means clustering and develop an explainable machine learning model for early prediction of at-risk students.

## 2. RESEARCH QUESTION

The central research question is:

*Can cluster-derived student behavioural profiles enable earlier and more interpretable at-risk prediction compared to raw-feature baselines?*

This decomposes into three sub-questions:

1. Which behavioural features from OULAD VLE logs most effectively distinguish distinct student engagement profiles?
2. Does profile-enriched input yield higher F1 and AUC-ROC than the raw-feature XGBoost baseline?
3. Do SHAP feature-importance rankings differ systematically across learner clusters, and which behavioural drivers are most predictive of at-risk outcomes within each profile?

### 3. LITERATURE REVIEW

#### 3.1. UNSUPERVISED LEARNER PROFILING

Clustering-based learner profiling has emerged as a principal strategy for converting raw VLE interaction data into interpretable student typologies. [Alzahrani et al. \(2025\)](#) engineered weekly LMS activity features and applied K-means clustering to segment 127 students into high, moderate, and low engagement profiles, achieving strong internal and label-based validity; however, the study covers a single course and does not extend to predictive modelling. [El Ghali et al. \(2025\)](#) applied hierarchical clustering with PCA to OULAD, producing four profiles at a silhouette score of 0.93, yet their feature set mixed demographic and outcome variables with engagement traces and the analysis stopped short of connecting profiles to downstream prediction or explainability. [Mdaghri-Alaoui et al. \(2025\)](#) ran parallel K-Means and Hierarchical Agglomerative Clustering on behavioural and demographic dimensions of OULAD separately, yielding profiles validated by classifiers above 0.99 accuracy, but the behavioural features were limited to average score and total clicks, omitting richer temporal VLE signals. Collectively, these studies confirm that unsupervised profiling on OULAD reliably uncovers distinct learner archetypes, but none couples profiling with downstream supervised prediction or cluster-level explanation.

#### 3.2. SUPERVISED ENGAGEMENT PREDICTION

Early work by [Al-Shabandar et al. \(2019\)](#) applied GBM and neural networks across Harvard MOOC and OULAD data, with GBM achieving 95.2% on a learning-achievement task using full-course VLE clickstream aggregates, establishing ensemble tree models as the strongest family for VLE-based prediction. [Torkhani and Rezgui \(2025\)](#) subsequently benchmarked six ML classifiers and four deep learning models on the OULAD AAA module (748 students); Random Forest led ML performance (accuracy 82.73%) while LSTM led overall (83.41%), yet the study covers one module with no clustering or explainability. [AlFaress et al. \(2025\)](#) evaluated five classifiers against a 44-feature set of VLE clickstream, assessment, and demographic variables on OULAD; Gradient Boosting achieved the highest performance (accuracy 91.02%, F1-score 91.18%), and the prominence of the *imd\_band* deprivation index as a top predictor flags an algorithmic-bias risk if deployed without a fairness audit. [Alalawi et al. \(2025\)](#) validate a prediction–intervention pipeline in a live tertiary setting (pass rates +35%), but rely solely on assessment scores without VLE traces or clustering. Most recently, [Mansouri et al. \(2026\)](#) achieved 90% accuracy on OULAD using a SMOTE–GA–GRU deep learning pipeline; however, the authors themselves note that GRU predictions are difficult to explain, motivating the use of interpretable tree-based models in the present project.

#### 3.3. EXPLAINABILITY IN EDUCATIONAL ML

Explainability techniques have become essential for translating black-box model outputs into actionable insights for educators and policymakers. [Ahmed et al. \(2025\)](#) applied both SHAP and LIME to the same ensemble voting regressor: global SHAP attribution revealed that prior academic scores accounted for approximately 62% of total model influence while study hours ranked second, and LIME provided localised instance-level explanations for individual predictions. The study relies, however, on generic Kaggle datasets that contain no longitudinal VLE clickstream data or learner-clustering structure, leaving open how feature importances shift across distinct engagement archetypes. [Islam et al. \(2025\)](#) similarly applied SHAP, LIME, and

ELI5 to an XGBoost model on a Portuguese dropout dataset, confirming curricular progression as the dominant predictor; yet their analysis remains global, with no VLE traces or cluster-stratified explanation. This gap motivates the present project’s use of SHAP applied per-cluster, enabling comparison of the key behavioural drivers of at-risk outcomes across distinct learner archetypes rather than across the population as a whole.

### 3.4. RESEARCH GAP

The absence of an integrated clustering–prediction–explainability pipeline has been noted across nearly a decade of learning analytics research. [Alhakbani and Alnassar \(2022\)](#) surveyed 18 OULAD benchmark studies up to 2022 and found model explanation consistently absent. [Jin et al. \(2024\)](#) subsequently conducted a PRISMA review of 17 OULAD-focused papers spanning 2017–2024 and likewise identified no work combining explainability with predictive modelling; [Torkhani and Rezgui \(2025\)](#) exemplify this pattern, achieving competitive prediction accuracy on OULAD without any clustering or explanation layer. More recent 2025 work has advanced individual threads without bridging them. Studies applying K-means to OULAD, such as [El Ghali et al. \(2025\)](#) and [Alzahrani et al. \(2025\)](#), construct behavioural profiles but stop short of feeding those profiles into a downstream supervised model — leaving the causal link between learner archetypes and predictive signal unexplored. Conversely, supervised prediction studies ([Torkhani and Rezgui, 2025](#); [AlFaress et al., 2025](#)) achieve strong accuracy but treat the student population as a homogeneous mass, neither grouping learners by behaviour nor providing cluster-specific explanations. Global SHAP attribution ([Ahmed et al., 2025](#); [Islam et al., 2025](#)) and prediction-to-intervention frameworks ([Alalawi et al., 2025](#)) both omit VLE behavioural clustering and per-cluster explainability. The present project directly addresses this compound gap by integrating K-means clustering on VLE interaction sequences with XGBoost-based early-warning prediction and per-cluster SHAP explainability on the OULAD dataset.

## 4. ANTICIPATED OUTCOMES

The project outcomes are structured into three tiers as shown in [Table 1](#).

## 5. PROJECT TIME PLAN

### 5.1. MILESTONE OVERVIEW

Key phases and deliverables are summarised in [Table 2](#).

### 5.2. GANTT CHART

The week-by-week schedule is shown in [Figure 1](#).

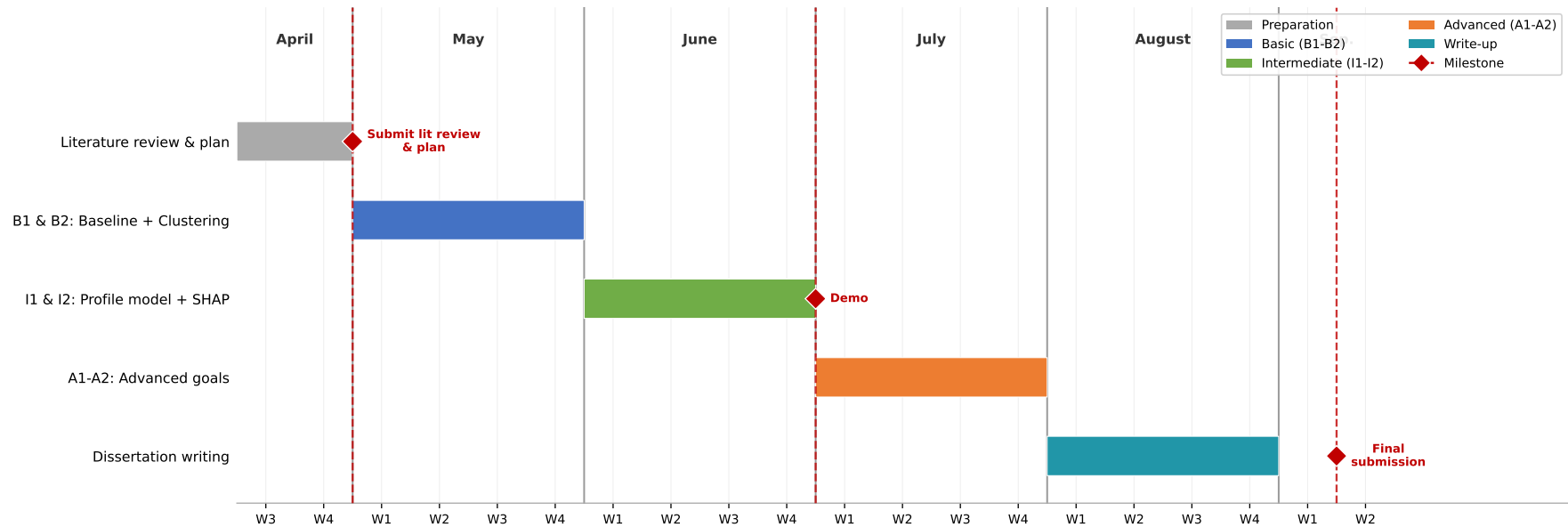
Table 1: Project goals by tier, with risk assessment.

| Tier         | ID | Description                                                                                                                                             | Risk   |
|--------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Basic        | B1 | Train raw-feature XGBoost baseline on OULAD; report Accuracy, Precision, Recall, macro-F1, AUC-ROC                                                      | Low    |
| Basic        | B2 | Feature engineering from VLE logs; apply k-Means; validate with silhouette score; visualise clusters (PCA / t-SNE) and interpret profiles educationally | Low    |
| Intermediate | I1 | Train profile-enriched XGBoost model (same architecture as B1, cluster label as additional feature); compare F1 and AUC-ROC to B1                       | Medium |
| Intermediate | I2 | Apply SHAP to both models; produce comparative interpretability analysis                                                                                | Medium |
| Advanced     | A1 | Cross-module profile transferability study                                                                                                              | High   |
| Advanced     | A2 | Multi-class engagement prediction                                                                                                                       | High   |

Table 2: Key milestones and deliverables.

| Phase                                    | Period                           | Key Deliverable                                                                 |
|------------------------------------------|----------------------------------|---------------------------------------------------------------------------------|
| Preparation                              | Apr W3 – Apr W4                  | Literature review & plan writing                                                |
| <b>Formative submit</b><br>Basic (B1–B2) | <b>Apr W4</b><br>May W1 – May W4 | <b>Submit lit review &amp; plan</b><br>XGBoost baseline; clustering & profiling |
| Intermediate (I1–I2)                     | Jun W1 – Jun W4                  | Profile model + SHAP analysis                                                   |
| <b>Demo</b><br>Advanced (A1–A2)          | <b>Jun W4</b><br>Jul W1 – Jul W4 | <b>Supervisor demo</b><br>Cross-module; multi-class prediction                  |
| Write-up                                 | Aug W1 – Aug W4                  | Dissertation draft & revisions                                                  |
| <b>Final submission</b>                  | <b>Sep W1 (Sep 4, 2026)</b>      | Report PDF + code + log                                                         |
| Oral exam                                | Week of Sep 14, 2026             | 30-min online oral exam                                                         |

Figure 1: Project Gantt chart. Dashed red lines and diamonds mark key deadlines; see legend for colour coding.



## 6. DEMONSTRATING SUCCESS

### 6.1. CLUSTERING QUALITY (ANSWERING Q1)

Sub-question Q1 — *which behavioural features most effectively distinguish distinct student engagement profiles?* — is answered by goal B2. The K-Means solution is evaluated by:

- **Silhouette Score:** Target  $> 0.4$ , indicating well-separated and cohesive clusters.
- **Davies–Bouldin Index:** Lower values indicate better inter-cluster separation; used as a secondary internal metric.
- **Educational interpretability:** Each cluster must correspond to a recognisable learner archetype (e.g., high-engagement, intermittent, disengaged), validated through per-cluster feature distributions and PCA/t-SNE visualisation.

### 6.2. QUANTITATIVE EVALUATION OF PREDICTIVE MODELS (ANSWERING Q2)

Sub-question Q2 — *does profile-enriched input yield higher F1 and AUC-ROC than the raw-feature XGBoost baseline?* — is answered by comparing goals B1 and I1 under a consistent experimental protocol:

- **Train/test split:** Stratified 80/20 hold-out with 5-fold cross-validation.
- **Metrics:** Accuracy, Precision, Recall, macro-F1, and AUC-ROC; Recall for the at-risk class is prioritised given the cost of false negatives.
- **I1 vs. B1:** McNemar’s test on paired prediction vectors determines whether I1 yields a statistically significant improvement in macro-F1 and AUC-ROC over B1; a positive result answers Q2 affirmatively.
- **Early-warning simulation:** Training data restricted to the first  $N$  weeks ( $N \in \{2, 4, 6\}$ ) to assess whether cluster profiles remain informative at earlier intervention windows.

### 6.3. INTERPRETABILITY ASSESSMENT (ANSWERING Q3)

Sub-question Q3 — *do SHAP feature-importance rankings differ systematically across learner clusters, and which behavioural drivers are most predictive of at-risk outcomes within each profile?* — is answered by goal I2:

- **Global SHAP rankings:** Compare feature-importance rankings between B1 and I1 to quantify the explanatory value added by the cluster label beyond raw VLE aggregates.
- **Per-cluster beeswarm plots:** Identify which behavioural features drive at-risk predictions differently across learner profiles, enabling cluster-stratified explanation rather than population-level attribution alone.

Together, the answers to Q1–Q3 address the central research question: Q1 confirms that clustering extracts valid behavioural profiles; Q2 tests whether those profiles improve prediction; and Q3 reveals the behavioural mechanisms behind the predictions.

## REFERENCES

- Wesam Ahmed, Mudasir Ahmad Wani, Pawel Plawiak, Souham Meshoul, Amena Mahmoud, and Mohamed Hammad. Machine learning-based academic performance prediction with explainability for enhanced decision-making in educational institutions. *Scientific Reports*, 15: 26879, 2025. doi:[10.1038/s41598-025-12353-4](https://doi.org/10.1038/s41598-025-12353-4).
- Raghad Al-Shabandar, Abir Jaafar Hussain, Panos Liatsis, and Robert Keight. Detecting at-risk students with early interventions using machine learning techniques. *IEEE Access*, 7: 149464–149478, 2019. doi:[10.1109/ACCESS.2019.2943351](https://doi.org/10.1109/ACCESS.2019.2943351).
- Khalid Alalawi, Rukshan Athauda, and Raymond Chiong. An extended learning analytics framework integrating machine learning and pedagogical approaches for student performance prediction and intervention. *International Journal of Artificial Intelligence in Education*, 35: 1239–1287, 2025. doi:[10.1007/s40593-024-00429-7](https://doi.org/10.1007/s40593-024-00429-7).
- Mahmoud Yousef AlFaress, Midhunchakkaravarthy Janarthanan, and Chandra Kumar Dixit. Early detection of at-risk students using machine learning: An empirical study on the OULAD with a focus on ethical feature selection. *SGS Engineering & Sciences*, 1(4), 2025. URL <https://spast.org/techrep/article/view/5692>.
- Haya A. Alhakbani and Fatema M. Alnassar. Open learning analytics: A systematic review of benchmark studies using open university learning analytics dataset (OULAD). In *Proceedings of the 2022 7th International Conference on Machine Learning Technologies (ICMLT)*, pages 81–86. ACM, 2022. doi:[10.1145/3529399.3529413](https://doi.org/10.1145/3529399.3529413).
- Nisreen Alzahrani, Maram Meccawy, Halima Samra, and Hassan A. El-Sabagh. Identifying weekly student engagement patterns in e-learning via k-means clustering and label-based validation. *Electronics*, 14(15):3018, 2025. doi:[10.3390/electronics14153018](https://doi.org/10.3390/electronics14153018).
- Mohamed El Ghali, Issam Atouf, Kamal El Guemmat, Said Broumi, and Mohamed Talea. Enhancing e-learning through strategic student segmentation: Insights from the OULAD database. *Journal of Theoretical and Applied Information Technology*, 103(4):1290–1299, 2025.
- Md. Mahmudul Islam, Farhad Hossain Sojib, Md. Fazle Hasan Mihad, Mahmudul Hasan, and Mahfujur Rahman. The integration of explainable AI in educational data mining for student academic performance prediction and support system. *Telematics and Informatics Reports*, 18:100203, 2025. doi:[10.1016/j.teler.2025.100203](https://doi.org/10.1016/j.teler.2025.100203).
- Lingxi Jin, Yao Wang, Huiying Song, and Hyo-Jeong So. Predictive modelling with the open university learning analytics dataset (OULAD): A systematic literature review. In *Artificial Intelligence in Education. Posters and Late Breaking Results, Workshops and Tutorials, Industry and Innovation Tracks, Practitioners, Doctoral Consortium and Blue Sky (AIED 2024)*, volume 2150 of *Communications in Computer and Information Science*, pages 477–484. Springer, Cham, 2024. doi:[10.1007/978-3-031-64315-6\\_46](https://doi.org/10.1007/978-3-031-64315-6_46).
- Jakub Kuzílek, Martin Hlosta, and Zdenek Zdrahal. Open university learning analytics dataset. *Scientific Data*, 4:170171, 2017. doi:[10.1038/sdata.2017.171](https://doi.org/10.1038/sdata.2017.171).

- Nesrine Mansouri, Mourad Abed, Makram Soui, and Mohammed Kutbi. On the use of advanced learning analytics in student at-risk prediction: A deep learning model. *Technology, Knowledge and Learning*, 2026. doi:[10.1007/s10758-025-09937-4](https://doi.org/10.1007/s10758-025-09937-4).
- Ghalia Mdaghri-Alaoui, Ilhame Khabbachi, Abdelhamid Zouhair, and El Mokhtar En-Naimi. Clustering student profiles with k-means and HAC for future adaptive virtual learning agents. In *2025 4th International Conference on Embedded Systems and Artificial Intelligence (ESAI)*. IEEE, 2025. doi:[10.1109/ESAI67033.2025.11438418](https://doi.org/10.1109/ESAI67033.2025.11438418).
- Wala Torkhani and Kalthoum Rezgui. OULAD MOOC student performance prediction using machine and deep learning techniques. In *Proceedings of the International Conference on Decision Aid and Artificial Intelligence (ICODAI 2024)*, volume 12 of *Atlantis Highlights in Intelligent Systems*, pages 229–241, 2025. doi:[10.2991/978-94-6463-654-3\\_18](https://doi.org/10.2991/978-94-6463-654-3_18).